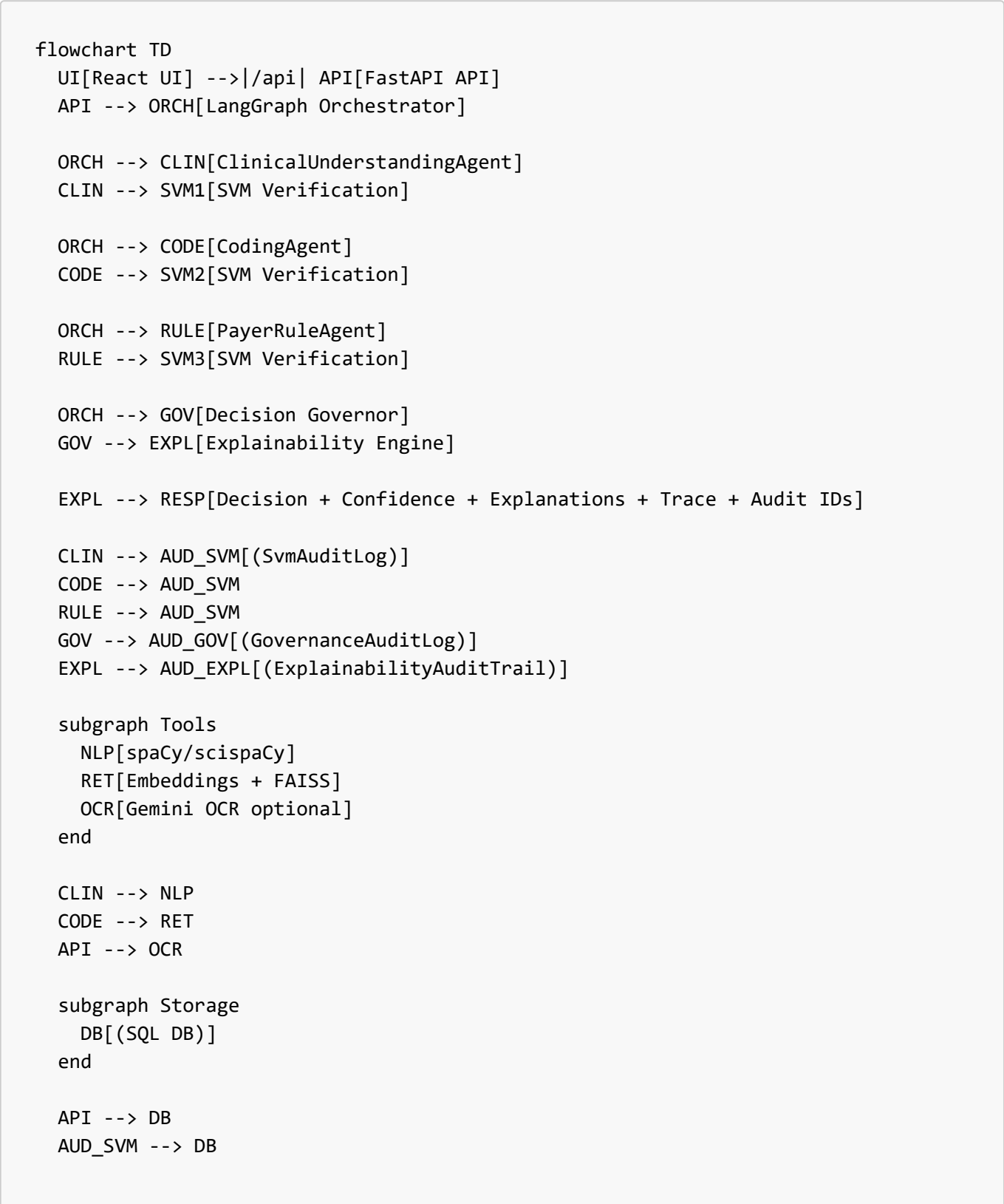


MedLedger AI — Architecture, Agent Roles, and Impact Model

This document provides a compact 1–2 page view of agent roles, how they communicate, tool integrations, and error-handling logic, plus the impact model used by the denial recovery dashboard.

System Diagram (Agents + Verification + Governance + Explainability)



```
AUD_GOV --> DB
AUD_EXPL --> DB
```

Agent Roles (Responsibilities)

- **ClinicalUnderstandingAgent**
 - Extracts structured clinical entities (diagnoses, procedures, medications) from unstructured notes.
 - Produces evidence-backed entities used downstream for coding and rules.
- **CodingAgent**
 - Suggests ICD-10 codes and supporting evidence (semantic similarity / matches).
 - Uses retrieval (embeddings/FAISS) to ground code suggestions.
- **PayerRuleAgent**
 - Evaluates payer/TPA rules against the claim context.
 - Emits violations, missing documentation requirements, and policy flags.
- **SVM Verification (per-stage)**
 - Runs after clinical, coding, and rules stages.
 - Produces stage scores (alignment, consistency, reasonability) and structured issues.
- **Decision Governor (Governance Layer)**
 - Aggregates confidence and applies thresholds/policy actions.
 - Outputs a final decision: **APPROVE** | **WARN** | **BLOCK** | **ESCALATE**, plus a final confidence.
- **Explainability Engine**
 - Renders human-readable explanations from config-driven templates and selection rules.
 - Outputs both readable explanations and structured details (template_id, rule_id, params) for audit/replay.

How They Communicate (State + Trace)

Communication between agents is structured and traceable:

- **Shared workflow state**
 - The orchestrator maintains a structured state object (entities, codes, rule results, verification scores, governance outputs).
 - Each agent reads the fields it needs and writes new fields for downstream agents.
- **Traceability**
 - Each run is traceable via a **trace_id** and ordered steps, allowing replay and audit.
 - Audit records link to the same run metadata so the pipeline is reconstructible.
- **Structured I/O**

- Agents exchange JSON-like structures rather than free-form text.
- Explanations are rendered at the end from templates and rules, not hardcoded strings.

Tool Integrations (What Gets Called)

- **spaCy/scispaCy**: clinical entity extraction from noisy clinical text.
- **Embeddings + FAISS**: semantic retrieval for coding assistance and matching.
- **Gemini (optional)**: fallback OCR/extraction for scanned/handwritten inputs (when enabled).

Error Handling Logic (What Happens When Things Go Wrong)

Error-handling is designed to prevent crashes and preserve auditability:

- **Stage errors become structured issues**
 - Parsing failures, missing signals, or low-quality extraction are surfaced as structured issues.
 - Downstream agents can continue (where safe) using partial outputs.
- **Verification downgrades confidence**
 - SVM verification can reduce confidence or mark stage outputs as unreliable.
 - Governance can escalate the decision based on verification issues.
- **Governance gating**
 - Guardrails can override earlier "success" and force **WARN/BLOCK/ESCALATE**.
- **Explainability resilience**
 - Explanation generation is template/rule-driven and can emit partial explanations if a single item fails to render.
 - The response always remains structured and audit-linked.

Denial Recovery Diagram (Email + Vapi Call + Corrections)

```

flowchart TD
    S[Claim Status Update: Denied] --> E[DenialEvent Created]
    E -->|if email exists| GMAIL[Gmail Ingestion + Parser]
    GMAIL --> FACTS[Denial Facts]
    E -->|if no email/details| VAPI[Vapi Outbound Call]
    VAPI -->|end-of-call-report webhook| WEBHOOK[FastAPI /denials/vapi/webhook]
    WEBHOOK --> FACTS
    FACTS --> AG[DenialManagementAgent]
    AG --> CORR[Corrections + Required Docs]
    AG --> RESUB[Resubmission Plan]
    AG --> LEARN[Outcome Learning]
  
```

Notes:

- If you run locally, Vapi webhooks require a public URL (ngrok). On localhost, use a tunnel or the manual “sync result” approach to pull call results.

Impact Model (Business KPIs)

The impact model converts denials, actions, and outcomes into metrics displayed on the Business Impact Dashboard (</impact>).

Impact Flow

```
flowchart TD
    A[DenialEvent] --> B[Denial Facts]
    B --> C[Corrections Proposed]
    C --> D[Resubmission Created]
    D --> E[Outcome Recorded]
    E --> F[Impact KPIs]
```

KPI Definitions

- **Revenue recovered (₹):** sum of recovered/paid outcomes after denial-driven corrections and resubmission.
- **Denial reduction (%):** reduction in denied outcomes vs total submissions over the same time window (demo uses recorded outcomes as a proxy).
- **Automation (%):** percentage of denial cases where the system produced an actionable correction/resubmission plan without manual override.
- **Time-to-resolution:** time from denial event creation to successful recorded outcome.

Evidence and Auditability

Each KPI is backed by per-claim evidence:

- denial reason/root cause (from email or call)
- correction actions and documents required
- resubmission record
- linked audit logs for explainability, governance, and verification